higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

# MARKING GUIDELINE

## NATIONAL CERTIFICATE (VOCATIONAL)

## COMPUTER PROGRAMMING
(First Paper)
**NQF LEVEL 4**

**27 NOVEMBER 2019**

**This marking guideline consists of 5 pages.**

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## SECTION A

## QUESTION 1

1.1 B
1.2 D
1.3 B
1.4 C
1.5 A
1.6 C
1.7 B
1.8 B
1.9 D
1.10 A

(10 × 1) **[10]**

## QUESTION 2

2.1 False
2.2 True
2.3 True
2.4 False
2.5 False
2.6 True
2.7 False
2.8 True
2.9 True
2.10 False

(10 × 1) **[10]**

## QUESTION 3

3.1 Normalisation
3.2 Polymorphism
3.3 Version control/Source management
3.4 Hyperlink
3.5 Software bug

(5 × 1) **[5]**

**TOTAL SECTION A: 25**

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## SECTION B

## QUESTION 4

4.1
- Reusable✓ – reuse same block of code again and again
- Extendible✓ – add new blocks of code to existing ones to suit needs
- Maintainable✓ – because OOP is modular it is much easier to read and understand (3)

4.2
- Test the functionality.
- Test the script on a number of commonly used browsers.
- Test for error handling or check for syntax errors. (Any relevant 3 × 1) (3)

4.3 4.3.1
- Customer
- Vehicle
- Service (Any relevant 3 × 1) (3)

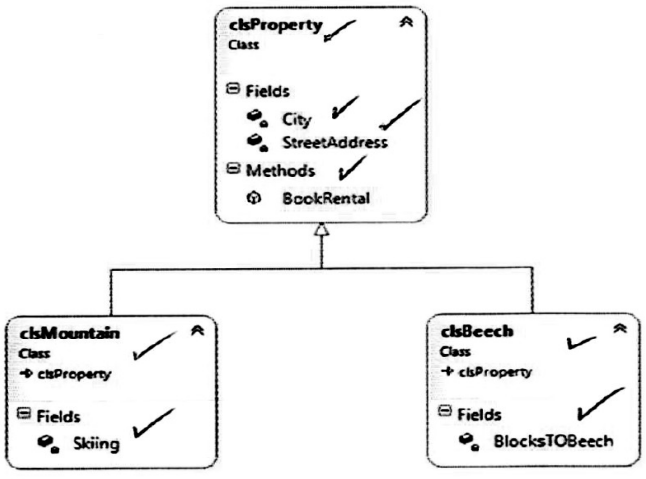
4.3.2
- Customer's attributes: Name,✓ contact number✓
- Customer's method: Display information✓
- Vehicle's attribute: Colour, Registration number,✓ Model✓
- Vehicle's method: Display Information✓
- Service's attribute: Description, Price✓
- Service's method: GetPrice,✓ CalcPriveExcVAT✓

(Any relevant 9 × 1) (9)

4.4 4.4.1 Advance HTML text editor with the intellisense feature
4.4.2 PHP and MYSQL, ASP.NET for scripting and data storage
4.4.3 Digital camera and photo-editing tool
4.4.4 Syntax-checking editor and browser which can provide script-error debugging facilities

(Any relevant 4 × 2) (8)

4.5




Superclass, hierarchy, mountain, beach (8)
**[34]**

## QUESTION 5

5.1 5.1.1 Used to find ✓ and or sort✓ data in a table (2)

5.1.2
- Primary key
- Foreign key (2)

5.2
- Web form
- Third-party applications
- Client-side or server-side scripting (3)

5.3 Bandwidth in kbps:
10 × 1 000 kbps = 10 000 kbps✓
Data in KB:
20 GB = 20 000 000 KB✓
Download time = (20 000 000 * 8) KB/10 000 kbps✓
Therefore, the user needs 16 000 seconds.✓
(Or any other valid/correct calculation) (4)

5.4
```
function✓ Validate()✓ {
  var value;✓
  value=frmForm1.txtContactNo.value;✓
  if✓ (value.Length < 1)✓{
     window.alert✓ ("Contact number field cannot be empty.");✓
  }
}
```

Also accept: if (value.length==0) (8)
**[19]**

## QUESTION 6

6.1 6.1.1 The server requests the username and password from the user✓ and then compares it with the ones stored in the database. ✓

6.1.2 It is used to strengthen the challenge response. ✓ Users get a once-off pin on their cellphone or via e-mail to complete the authentication process. ✓

6.1.3 It converts data to an unrecognisable or encrypted form to make it unreadable for unauthorised viewers.✓ It is also used to secure data across HTTPS and SSL Internet connections.✓

(3 × 2) (6)

6.2 6.2.1 In 1:1, one occurrence of an entity type relates to only one occurrence of the other entity type,✓ e.g. one lecture teaches CP L4 group 1 and CP L4 group 1 is taught by one lecturer.✓

(Any relevant answer) (2)

6.2.2 In 1:M, one occurrence of an entity type relates to one or many occurrences of the other entity type,✓ e.g. one tutor supervises one or more students and many students are supervised by one tutor.✓ (Any relevant answer) (2)

6.3 6.3.1 The webpage download time is very slow and delay is very high. Implications:✓ Limits the size of webpages and removes or reduces the number of interactive webpages.✓ (2)

6.3.2 The webpage download time is very good and delay is very low. Implications:✓ Highly interactive webpages with audio and/or video links✓ (2)

6.4

```
<form>✓
<fieldset>✓<legend>Customer Information</legend>✓
  <label>Name:</label> <input type="text" name="txtName">✓<br><br>
  <input type="radio" name="gender">Male✓
  <input type="radio" name="gender">Female✓
  <input type="radio" name="gender">Other<br><br>✓
  <input type="button" value="Submit" name="">✓
</fieldset>
</form>
```

(8)

**[22]**

**TOTAL SECTION B: 75**
**GRAND TOTAL: 100**

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